

Title

AI Circuit Breaker

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Abstract

Trust in engineered systems is usually accompanied by calibration certificates and stated uncertainty budgets. For example, ECG monitors can be trusted to assess human heart health because their signals trace to measurement standards and their error bounds are known. AI systems offered to do similar safety-critical work typically come with neither. Systems engineering (SE) already supplies this kind of measurement rigor to biomedical devices, aerospace, and manufacturing. The discipline is called metrology: the engineering of measurement with stated uncertainty. AI is the exception: calibration practice is minimal, uncertainty budgets are rarely stated, and traceability to reference standards is limited.

The existing AI-trust literature contributes three strands: policy and standards frameworks, assurance-case templates and runtime monitors, and uncertainty-quantification surveys. None, to our understanding, combines these into a single continuous calibrated measurement with a traceable reference and stated uncertainty inside the SE venue.

SE is being asked to field AI faster, not slower. Without a measurement basis, the speed-versus-safety tradeoff is unpriced: teams either move slowly, waiting for confidence they cannot quantify, or move fast and absorb unquantified risk.

We treat AI trustworthiness as a measurement problem built from five research traditions: *systems-theoretic morphisms* (structure-preserving mappings between a model and what it models); the *Guide to the Expression of Uncertainty in Measurement* (GUM) [JCGM 100:2008]; *statistical process control* (SPC) [Montgomery]; *five-dimensional context relevancy* (WHO/WHAT/WHEN/WHERE/WHY) grounded in structured assurance levels; and *Systems-Theoretic Process Analysis* (STPA/STAMP) [Leveson], which treats safety as a control problem rather than a failure-probability problem. The AI Circuit Breaker extends two lines of our prior work: the pairing of Bayesian methods with systems theory for test and evaluation of learning-based systems [Wach et al., INSIGHT 2022], and the isomorphic-degradation framework for morphism quality [Wach et al., CSER 2026]. These precursors are composed here into a single instrument with four stated objectives.

Objectives

Objective 1: Establish morphism quality as the trust measurand. The primary measurand is morphism quality: how well the AI's internal picture of the world matches the world itself. The two-axis morphism distance (D_s structural, D_b behavioral) is extended by a third axis, context relevancy distance (D_r), which captures whether the right slice of reality is being modeled in the first place. An AI can be near-isomorphic to the wrong context—low D_s , low D_b —and still be catastrophically wrong for the situation at hand. Measurement supplies the evidence base for human judgment, not the decision.

Objective 2: Build an enforcement architecture. Turn the three-axis measurement into action, not just a dashboard readout, grounded in causal hazard structure rather than statistical exceedance alone.

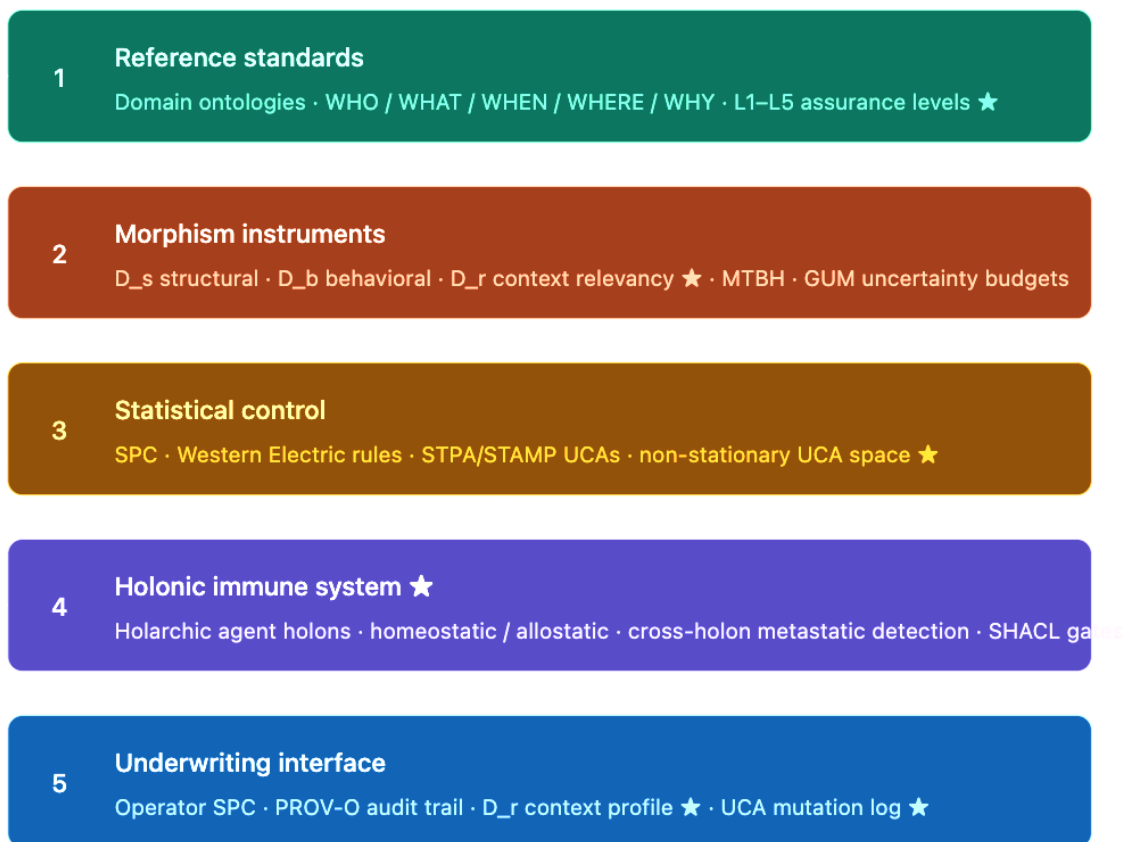
Objective 3: Ground enforcement in causal hazard structure. Extend the measurement architecture with STPA/STAMP-derived unsafe control action (UCA) detection, including detection of non-stationary UCA spaces that emerge when learning pipelines and human-in-the-loop (HiTL) feedback operate at the outer recursive tiers of a dynamically evolving control hierarchy.

Objective 4: Add a bio-inspired holonic regulation layer. Prevent the instrument from degrading silently over time and resolve the domain-transfer scaling problem through a holonic architecture that provides a native principle for self-similar deployment across domains and scales, extending Layer 4 as a federated immune system capable of detecting metastatic adaptation patterns across agent instances.

In combination, these four objectives compress the 'when is it safe to field' decision cycle, because trust becomes priceable in units that decision-makers can use.

Methodology: A Five-Layer Measurement Architecture

The AI Circuit Breaker is a five-layer measurement architecture. Each layer performs a specific metrological function, and the feedback arrows between layers encode the closed-loop logic that keeps the instrument calibrated and the operator informed.



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Layer 1, Reference Standards, holds calibrated specifications of valid domain behavior. Domain ontologies serve as gauge blocks: version-controlled, uncertainty-bounded, and recalibrated on schedule. Layer 1 now also anchors **five-dimensional context relevancy**: WHO (agent identity and authorization scope), WHAT (operational boundary), WHEN (temporal validity window), WHERE (deployment context), and WHY (mission intent). Each dimension maps to a structured assurance level (L1–L5), providing the reference against which the operational context of any given AI invocation is compared. This is the basis for measuring D_r : without it, morphism quality has no mechanism to detect that a well-calibrated AI has been invoked outside the envelope for which it was calibrated.

Layer 2, Morphism Instruments, measures AI-reality disagreement on three axes: structural distance (D_s), behavioral distance (D_b), and context relevancy distance (D_r). D_s and D_b measure how well the AI models a given slice of reality; D_r measures whether that slice is the right one. An AI system can exhibit low D_s and low D_b —near-isomorphic to its training context—while remaining catastrophically misaligned with the operational context at hand. D_r captures this gap by comparing the live five-dimensional relevancy profile against the L1–L5 anchors in Layer 1. MTBH (mean time between hallucinations) summarizes longitudinal reliability across all three axes. Each instrument carries a GUM-style uncertainty budget separating statistical (Type A) from prior-knowledge (Type B) sources.

Layer 3, Statistical Control, turns measurements into verdicts. Thresholds are data-derived through SPC rather than guessed; Western Electric rules detect drift across all three morphism axes; a graduated response ranges from normal autonomy through caution, restriction, and halt to full human takeover. Layer 3 now incorporates an **STPA / STAMP causal hazard model**: thresholds are not only data-derived but grounded in unsafe control actions (UCAs) identified within the system's hierarchical control structure. Critically, when learning pipelines and HiTL feedback operate at the outer tiers of a recursive control hierarchy, the UCA space itself becomes **non-stationary**—the control structure can mutate. Layer 3, therefore, includes a UCA-space drift detector that monitors whether the operative control structure has changed shape, not merely whether outputs have drifted. To our knowledge, no existing SE or AI-safety framework addresses non-stationary UCA spaces in dynamically learning systems; this constitutes a distinct contribution to the STPA/STAMP literature and a direct collaboration opportunity with foundational work in systems-theoretic safety.

Layer 4, Closed-Loop Maintenance, is restructured as a **holonic immune system**. The holon—a self-contained whole that simultaneously participates as a component in a larger whole—provides the native scale principle that the architecture previously lacked. Each deployed AI agent instance is treated as a holon: carrying its own morphism instruments, SPC state, UCA boundary, and ontology version. These holons are embedded in a holarchy of agents operating at higher organizational scales, whose behavior changes with holarchic position just as a liver cell and a neuron share DNA yet perform different functions under different organizational constraints.

This holonic structure enables a bio-inspired immune system: homeostatic control maintains the operating envelope within each holon; allostatic control pre-adapts to high-risk windows across the holarchy; and cross-holon surveillance detects metastatic behavior—recursive adaptation patterns that propagate across agent instances in ways no individual holon's SPC would flag in isolation. Learning pipeline updates are gated by SHACL and SPARQL validation against Layer 1 at every holarchic level, preventing flagged failures from propagating upward through the holarchy. The holonic architecture also resolves the domain-transfer question posed in Phase III: porting to a new domain becomes a matter of instantiating a new holon with domain-specific ontology and baseline SPC data, rather than rebuilding the full instrument from scratch.

Layer 5, Underwriting Interface, presents the evidence to humans. Operator dashboards show SPC chart state, MTBH reliability, D_r context-relevancy status, and UCA-space stability indicators. A PROV-O (W3C Provenance Ontology) provenance model produces an audit trail queryable via SPARQL,

giving regulators access to the provenance of any single decision, including the five-dimensional context profile and any UCA-space mutations that preceded it.

Information decides a chain of morphisms (physical system through sensor, digital signal, processing, features, to final decision), with chain-level bounds for morphism distance developed in [Wach et al., CSER 2026]. The formal semantic backbone is specified in the work-in-progress Circuit Breaker Trust Ontology (CBTO), an OWL 2 DL design with 25 classes, 20 object properties, 12 data properties, 6 SHACL validation shapes, and 10 SPARQL competency queries, plus PROV-O provenance. The ontology is being extended to accommodate the five-dimensional relevancy framework, UCA-space state representation, and holarchic agent identity.

Phased Application Spectrum

We plan three phases on safety-critical classification testbeds, each testing a specific claim of the framework.

Phase I: Single-agent morphism monitoring. Prove that morphism quality can be measured continuously on one working AI classifier in a safety-critical domain. Scope includes one learned classifier, one domain ontology with five-dimensional L1–L5 anchors, and a baseline comparison against a conventional classifier operating without the Circuit Breaker layer.

Phase II: End-to-end composition, UCA integration, and bio-inspired regulation. Extend monitoring to the full observation chain (classification through risk alert to safe-state), demonstrate that chain-level composition bounds hold in practice, validate the UCA-space drift detector against known control structure mutations, and add the holonic regulation layer. Cross-holon metastatic pattern detection is verified against synthetic adversarial adaptation scenarios.

Phase III: Holonic domain-transfer protocol. Rather than simply estimating the cost to port the framework to a second safety-critical domain, Phase III executes the holonic instantiation protocol—instantiating a new holon with domain-specific ontology and baseline SPC data—and measures the actual effort against the Phase I baseline. The holonic architecture converts a cost-estimation exercise into a validated reuse protocol.

Expected Outcomes

Phase I targets: hallucination true-positive rate above 95% at 5% false-positive rate; classification performance at or above domain state-of-the-art; cold-start graduation within 25 SPC subgroups; end-to-end decision latency under 25 milliseconds; and D_r correctly flagging at least two context-mismatch scenarios that D_s/D_b alone would miss.

Phase II tightens these and adds: hallucination detection at 97%; SPC-based adversarial robustness above 95%; empirical confirmation of composition bounds across the full pipeline; at least one confirmed UCA-space mutation detected before an output-level SPC violation would have triggered; and at least one cross-holon metastatic pattern detected and contained.

Phase III delivers a validated holonic instantiation protocol for a second safety-critical domain, with empirical effort data that enterprises can apply to adoption decisions. The traceability chain, GUM-style uncertainty budgets, five-dimensional relevancy specification, UCA-space state representation, and work-in-progress CBTO are standalone outputs in their own right.

Relevance to Practice

SPC and metrology are already the vocabulary of regulated industries: biomedical, aerospace, and manufacturing. The Circuit Breaker translates AI safety into their existing mental models rather than asking them to learn new AI-safety jargon. Speed comes from letting people use tools they already know.

Three practical consequences follow. Engineers gain a decision-support instrument that expresses trust as a confidence interval rather than a binary pass or fail, with context relevance explicitly priced. Regulators gain a PROV-O audit trail that makes every AI decision's provenance queryable—including what context profile was active and whether the control structure had mutated. Program managers gain a cold-start protocol that ends deployment-readiness guesswork, and a holonic instantiation protocol that makes domain transfer a reproducible engineering activity rather than a project-level estimate.

The framework aligns with the Visibility and Analytics pillar of Zero Trust architectures [NSA, 2024], situates within the INCOSE AI4SE research roadmap [McDermott et al., INSIGHT 2020], and extends its test-and-evaluation and runtime-assurance directions.

Key Advancements

The primary contribution is the composition of five research traditions—systems-theoretic morphisms, GUM-based measurement uncertainty, statistical process control, five-dimensional context relevancy, and STPA/STAMP causal hazard modeling—into a single measurement instrument for AI trustworthiness, structured as a holonic immune system and situated inside the SE venue where, to our understanding, this combination has not previously appeared. Specific advancements include:

- **Three-axis morphism distance** (D_s structural, D_b behavioral, D_r context relevancy), where D_r maps five-dimensional WHO/WHAT/WHEN/WHERE/WHY profiles to L1–L5 assurance levels, detecting near-isomorphism to the wrong operational context—a failure mode invisible to the two-axis framework alone
- **Non-stationary UCA space detection via STPA/STAMP integration in Layer 3:** when learning pipelines and HiTL feedback operate at the outer tiers of a recursive control hierarchy, the UCA space can mutate; to our knowledge, this is the first SE framework to address this gap explicitly
- **Holonic immune system architecture for Layer 4:** each deployed agent instance is a self-calibrating holon within a federated holarchy, enabling cross-scale surveillance, metastatic adaptation detection, and domain portability through holonic instantiation rather than full re-implementation
- **Composition result for observation-mediated morphism chains**, extending the two-axis bounds from [Wach et al., CSER 2026] to three axes with five-dimensional relevancy bounds
- **GUM-style uncertainty propagation** applied end-to-end to AI trust measurement, now including D_r uncertainty
- **SPC-derived control limits** replacing intuited thresholds, paired with a defined cold-start protocol and UCA-grounded threshold derivation
- **CBTO formal semantic backbone** (OWL 2 DL + SHACL + SPARQL + PROV-O) extended for five-dimensional relevancy, UCA-space state, and holarchic agent identity

The result is speed and innovation delivered together: speed, because trust measurement turns deployment-readiness into a priceable engineering decision; innovation, because SE metrology now extends to AI components—with context relevancy, causal hazard grounding, and holonic scale—as it already extends to every other subsystem a safety-critical program fields.

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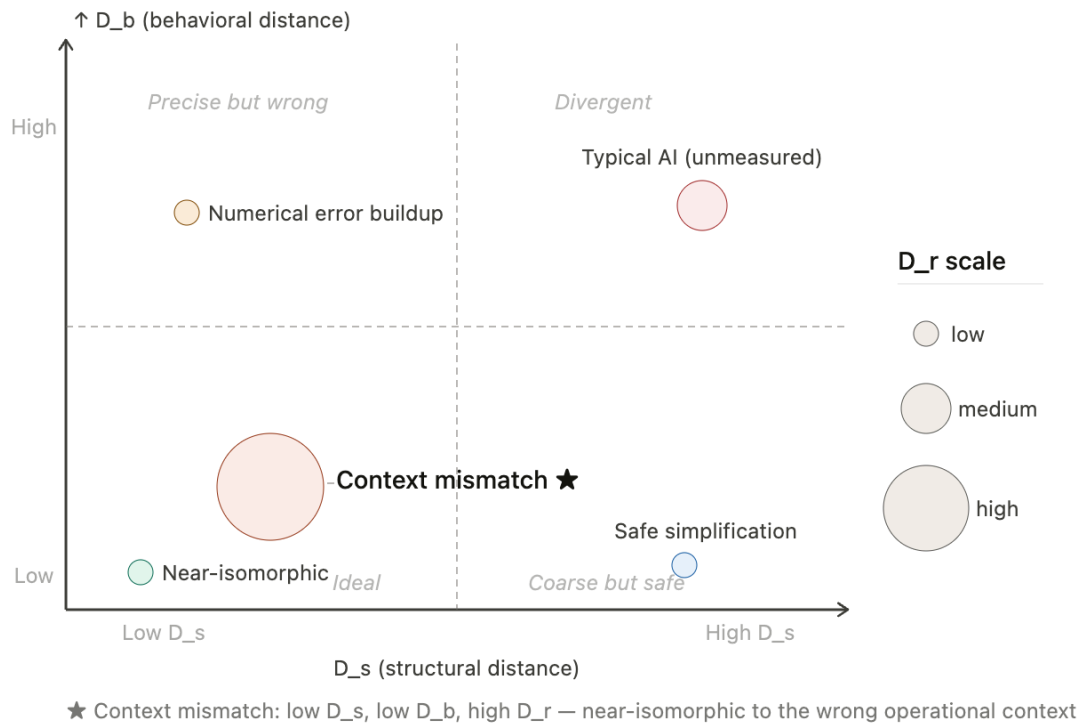


Figure 2: Two-axis morphism distance. D_s (structural distance, x-axis) captures how many distinct real-world states the AI lumps together. D_b (behavioral distance, y-axis) captures worst-case output discrepancy. Both are zero at an exact isomorphism. Four reference points illustrate typical positions: near-isomorphic (Ideal), numerical error buildup (Precise but Wrong; “discretization failure” in the formal literature), unmeasured AI (Divergent), and safe simplification (Coarse but Safe; “lumped-parameter model” in the formal literature).